

Economics 102
Homework #2
Due: February 1st at the beginning of class

Complete **all** of the problems. Please do **not** write your answers on this sheet. Show all of your work.

1. Suppose people only consume 3 different goods. The following table shows the prices and quantities of each good consumed in 2006, 2007, and 2008.

Year	Price of Fish	Quantity of Fish	Price of Pork	Quantity of Pork	Price of Beef	Quantity of Beef
2006	\$7	400	\$8	225	\$10	175
2007	8	550	7	250	12	275
2008	9	900	6	275	15	275

- a. Calculate nominal GDP in each of the three years.

Nominal GDP is simply equal to the sum of the current year price * current year quantity of all the goods.

$$2006: (7 \times 400) + (8 \times 225) + (10 \times 175) = 2,800 + 1,800 + 1,750 = \mathbf{\$6,350}.$$

$$2007: (8 \times 550) + (7 \times 250) + (12 \times 275) = 4,400 + 1,750 + 3,300 = \mathbf{\$9,450}.$$

$$2008: (9 \times 900) + (6 \times 275) + (15 \times 275) = 8,100 + 1,650 + 4,125 = \mathbf{\$13,875}.$$

- b. Calculate Real GDP in each of the three years, using 2006 as the base year.

Real GDP is equal to the sum of the base year price * current year quantity of all the goods.

$$2006: (7 \times 400) + (8 \times 225) + (10 \times 175) = 2,800 + 1,800 + 1,750 = \mathbf{\$6,350}.$$

$$2007: (7 \times 550) + (8 \times 250) + (10 \times 275) = 3,850 + 2,000 + 2,750 = \mathbf{\$8,600}.$$

$$2008: (7 \times 900) + (8 \times 275) + (10 \times 275) = 6,300 + 2,200 + 2,750 = \mathbf{\$11,250}.$$

- c. Calculate the GDP deflator for each of the three years.

The GDP deflator is equal to $(\text{Nominal GDP} / \text{Real GDP}) \times 100$.

2006: **100**. Because 2006 is the base year we know the deflator has to equal 100 even without doing any calculations.

$$2007: (9,450 / 8,600) \times 100 = \mathbf{109.9}.$$

$$2008: (13,875 / 11,250) \times 100 = \mathbf{123.3}.$$

- d. Calculate inflation for 2007 and 2008.

Inflation is equal to the growth rate of the GDP deflator. The growth rate formula is: $((\text{Year2} - \text{Year1}) / \text{Year1}) \times 100$.

$$2007: ((109.9 - 100) / 100) \times 100 = \mathbf{9.9\%}.$$

2008: $((123.3 - 109.9)/109.9)*100 = 12.2\%$.

e. Calculate Real GDP for 2007 and 2008 using the chain-weighted method.

Using 2006 as the base year, we know that Real GDP is equal to nominal GDP. Thus Real GDP in 2006 is \$6,350. This gives us the starting point for the chain-weighted method of calculating real GDP.

To calculate chain-weighted Real GDP for 2007 we need the following four pieces of information:

2006 quantities at 2006 prices: See part a, \$6,350.

2007 quantities at 2006 prices: See part b, \$8,600.

2006 quantities at 2007 prices: $(8*400) + (7*225) + (12*175) = \$6,875$.

2007 quantities at 2007 prices: See part a, \$9,450.

Now we calculate the growth rate of GDP with 2006 prices:

$((8,600 - 6,350)/6,350)*100 = 35.4\%$,

Then the growth rate of GDP using 2007 prices:

$((9,450 - 6,875)/6,875)*100 = 37.5\%$.

The next step is to average the two growth rates: $(35.4 + 37.5)/2 = 36.45\%$.

This gives us the chain weighted growth rate of real GDP for 2007. So to calculate 2007 Real GDP we multiply 2006 real GDP by this growth rate: $(6,350 + (6,350*36.45\%)) = \mathbf{\$8,664.6}$.

Repeating the same process for 2008 gives us the following:

2007 quantities at 2007 prices: See part a, \$9,450.

2008 quantities at 2007 prices: $(8*900) + (7*275) + (12*275) = \$12,425$.

2007 quantities at 2008 prices: $(9*550) + (6*250) + (15*275) = \$10,575$.

2008 quantities at 2008 prices: See part a, \$13,875.

Growth rate with 2007 prices: $((12,425 - 9,450)/9,450)*100 = 31.5\%$

Growth rate with 2008 prices: $((13,875 - 10,575)/10,575)*100 = 31.2\%$

Average growth rate: $(31.5\% + 31.2\%)/2 = 31.35\%$.

We use the chain-weighted 2007 GDP as our starting point, and get:

$(8,664.6 + (8,664.6*31.35\%)) = \mathbf{\$11,381.0}$.

f. Calculate the GDP deflator and inflation using Real GDP from part e.

The GDP deflator is still $(\text{Nominal GDP}/\text{Real GDP}) \times 100$ but now we are using the chain-weighted measure of Real GDP.

GDP deflator

2006: 100. 2006 is still the base year, so the GDP deflator is still 100.

2007: $(9,450/8,664.6) \times 100 = \mathbf{109.1}$.

2008: $(13,875/11,381.0) \times 100 = \mathbf{121.9}$.

Inflation

2007: $((109.1 - 100)/100) \times 100 = \mathbf{9.1\%}$.

2008: $((121.9 - 109.1)/109.1) \times 100 = \mathbf{11.7\%}$.

g. Are your answers to part d and part f the same? Why or why not?

Your answers are not the same. They are close, but there are significant differences. The first thing that we note is that the chain-weighted method gives us a lower measure of inflation. In 2007, the chain-weighted method measures inflation almost 1% lower than the standard method, while at the same time measuring Real GDP as a higher number. So under the chain-weighted method, real GDP growth is higher, and inflation is lower.

This is not always the case. The example in your book shows a case where using the chain-weighting method lowers GDP and raises inflation. The difference between the two methods is that the original method only uses base year prices. This means that your choice of base year actually matters. In the chain-weighted method, because you are using two years worth of prices to calculate growth rates, the base year does not affect the measurement of inflation. To check this, calculate inflation in 2008 using 2007 as the base year using both methods. I will put the results below, but you should verify my calculations.

This table measures inflation for 2008 using the two different methods of calculating real GDP.

	<u>2006 Base Year</u>	<u>2007 Base Year</u>
Standard method:	12.2%	11.7%
Chaining method:	11.7%	11.7%

h. Using the quantities from 2006 for your market basket, and 2006 as your base year, calculate the CPI for 2006, 2007 and 2008.

To calculate CPI, First: Fix the market basket. This means that the quantities you choose should not change for this part. For part a the fixed basket is: 400 fish, 225 pork and 175 beef. Step two is to find prices. The prices are given in the problem, and the prices will change. Step three is to compute the cost

of the market basket:

2006: $(\$7 \times 400) + (\$8 \times 225) + (\$10 \times 175) = 2800 + 1800 + 1750 = 6350$.
2007: $(\$8 \times 400) + (\$7 \times 225) + (\$12 \times 175) = 3200 + 1575 + 2100 = 6875$.
2008: $(\$9 \times 400) + (\$6 \times 225) + (\$15 \times 175) = 3600 + 1350 + 2625 = 7575$.
(Note: When calculating market basket cost for CPI *only* the prices change.)

Once we have the cost of the market basket, step four is to calculate CPI.

The formula for CPI is: $(\text{Current year cost} / \text{Base year cost}) \times 100$

2006 CPI: $(6350/6350) \times 100 = \mathbf{100}$.

2007 CPI: $(6875/6350) \times 100 = \mathbf{108.3}$.

2008 CPI: $(7575/6350) \times 100 = \mathbf{119.3}$.

i. Using the CPI calculate inflation.

The way that CPI is determined makes the inflation rate simply the growth rate of CPI. The growth rate formula is: $((\text{Year2} - \text{Year1}) / \text{Year1}) \times 100$.

2007 Inflation: $((108.3 - 100) / 100) \times 100 = \mathbf{8.3\%}$.

2008 Inflation: $((119.3 - 108.3) / 108.3) \times 100 = \mathbf{10.1\%}$.

j. Recalculate CPI and inflation using the 2006 quantities for your market basket but with 2008 as your base year.

When you change the base year, there is no need to recalculate the market basket cost, because the basket has not changed. You simply need to recalculate CPI.

2006 CPI: $(6350/7575) \times 100 = \mathbf{83.8}$.

2007 CPI: $(6875/7575) \times 100 = \mathbf{90.8}$.

2008 CPI: $(7575/7575) \times 100 = \mathbf{100}$.

Then recalculate inflation:

2007 Inflation: $((90.8 - 83.8) / 83.8) \times 100 = \mathbf{8.3\%}$.

2008 Inflation: $((100 - 90.8) / 90.8) \times 100 = \mathbf{10.1\%}$.

(Note: These inflation numbers are exactly the same as the ones in part i.)

k. Now calculate CPI and inflation using 2008 quantities as your market basket and 2006 as your base year

In this part the market basket has changed, so we need to recalculate the cost. Remember when calculating the market basket cost, the quantities remain fixed.

2006: $(\$7 \times 900) + (\$8 \times 275) + (\$10 \times 275) = 6300 + 2200 + 2750 = 11250$.

2007: $(\$8 \times 900) + (\$7 \times 275) + (\$12 \times 275) = 7200 + 1925 + 3300 = 12425$.

2008: $(\$9 \times 900) + (\$6 \times 275) + (\$15 \times 275) = 8100 + 1650 + 4125 = 13875$.

The CPI calculation does not change:

2006 CPI: $(11250/11250)*100 = 100$.
2007 CPI: $(12425/11250)*100 = 110.4$.
2008 CPI: $(13875/11250)*100 = 123.3$.

Recalculate inflation:

2007 Inflation: $((110.4-100)/100)*100 = 10.4\%$.

2008 Inflation: $((123.3-110.4)/110.4)*100 = 11.7\%$.

(Note: These numbers are very different than parts i and j, the difference is significant and not due to rounding error.)

- l. Does which year you choose as a base year matter? How about your choice of quantities for the market basket?

As you can see from the inflation numbers in parts i and j, the choice of a base year does not affect inflation at all. As a matter of fact the choice of a base year is completely arbitrary. However, changing the composition of the market basket is important. Notice that not all of the prices in this problem increased by the same amount over time. The price of fish increased by 28.6% $((9-7)/7)*100$, while the price of pork decreased by 25% $((6-8)/8)*100$ and the price of beef increased by 50% $((15-10)/10)*100$. Inflation would be different if the market basket contained only beef, then if the market basket contained only fish. So the choice of a market basket is very important in calculating inflation.

- m. Compare your answers from part d with your answers from part i. If they are the same, why are they the same? If they are different, why are they different?

The rate of inflation is different depending on the method that you use to calculate it. Note however that the numbers are similar. With the GDP deflator we are keeping prices fixed and changing quantities, with CPI we are keeping quantities fixed and changing prices.

2. Explain under what circumstances you would prefer to use the CPI measure of inflation and under what circumstances you would prefer the GDP deflator. Include a discussion of the pros and cons of both methods.

Since the CPI only measures consumer goods, it is only useful for consumer applications. Wages should be indexed by CPI because what we care about is allowing people to purchase the same basket of goods, likewise for any cost of living adjustment (COLA). One example of a COLA is with Social Security. The COLA was instituted to keep benefits at a constant level of purchasing power over time. Also anytime you wanted to measure the effect of imports, you need to use the CPI because the GDP deflator does not include imports in its calculation.

The GDP deflator would be more useful for manufacturing applications. The GDP deflator takes all aspects of the domestic economy into account, including the effects of plants, equipment and inventories. The CPI only measures consumer goods. Also if we are interested in comparing across countries, it may be more useful to use the GDP deflator.